Type MS · Explain how biceps, triceps and tendons move the arm

Answer 1 of 9 · Question 6 part c · 4 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>(cardiac muscle)
• striated/striped;
• branched;
• single/central nucleus;
(voluntary muscle)
• striated/striped;
• unbranched;
• many nuclei;
(involuntary muscle)
• non-striated/striped;
• unbranched;
• single/central nucleus;
• tapered at end;</td><td></td><td>max of 2 for each muscle
6</td></tr>
<tr><td></td><td>(ii)</td><td>small intestine/digestive tract/oesophagus/liver/uterus/arteries/veins/iris/sphincter muscles;</td><td>allow any named part of digestive tract</td><td>1</td></tr>
<tr><td>(b) (i)</td><td></td><td>1540;
(÷ 20) 77;</td><td>ECF incorrect addition</td><td>2</td></tr>
<tr><td></td><td>(ii)</td><td><table><tr><th>Percentage muscle mass (%)</th><th>Tally/frequency</th></tr><tr><td>70</td><td>ll 2</td></tr><tr><td>72</td><td>ll 2</td></tr><tr><td>74</td><td>llll 4</td></tr><tr><td>75</td><td>II 2</td></tr><tr><td>78</td><td>III 3</td></tr><tr><td>82</td><td>~~llll~~ l 6</td></tr><tr><td>84</td><td>l 1</td></tr></table></td><td>1 mark for suitable table, 1 mark for column headings and 1 mark for correctly inputted data and tally
allow just numbers in second column
accept just percentage/% as column heading</td><td>3</td></tr>
<tr><td></td><td>(iii)</td><td>82;</td><td>ECF from (ii)</td><td>1</td></tr>
<tr><td>(c)</td><td></td><td>• biceps contract;
• triceps relax;
• antagonistic pairs/work antagonistically;
• lower arm/radius pulled upwards;</td><td></td><td>4</td></tr>
</table>

Total Marks 17

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 1 of 9 · Question 6 part c · 4 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 2 of 9 · Question 1 part c · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

1	(a)	(i)	X cartilage; Y femur; Z knee cap;		1 1 1
		(ii)	**B** ligaments hold the leg bones together A is incorrect as ligaments do not attach muscles to bones C is incorrect as ligaments do not move leg bones D is incorrect as ligaments do not stop the leg bones rubbing together		1
		(iii)	hinge/synovial (joint)		1
	(b)		In the following order only oxygen; carbon dioxide;		 1 1
	(c)		biceps contracted/fatter; triceps relaxed/thinner; 		1 1

Total question 1 = 9 marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 2 of 9 · Question 1 part c · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 3 of 9 · Question 7 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

7 (a) (i)		aerobic respiration (1)		1
	(ii)	• shorter (1) • fatter/wider/thicker (1)		2
	(iii)	• shortening causes tension/pull (1) • tendon inelastic (1) • pulls on radius/bone (1)		3
(b)		• ball and socket indicated labelled bone (1) • capsule/ligament labelled (1) • cartilage labelled (1) • synovial fluid labelled (1) • synovial membrane labelled (1) • overall appearance of diagram (1)		6

Total 12 marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 3 of 9 · Question 7 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 4 of 9 · Question 7 part a.iii · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

7 (a) (i)		aerobic respiration (1)		1
	(ii)	• shorter (1) • fatter/wider/thicker (1)		2
	(iii)	• shortening causes tension/pull (1) • tendon inelastic (1) • pulls on radius/bone (1)		3
(b)		• ball and socket indicated labelled bone (1) • capsule/ligament labelled (1) • cartilage labelled (1) • synovial fluid labelled (1) • synovial membrane labelled (1) • overall appearance of diagram (1)		6

Total 12 marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 4 of 9 · Question 7 part a.iii · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 5 of 9 · Question 6 part d · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

6 (a)	• where two bones; • come together/meet;		2
(b) (i)	• humerus; • ulna;		2
(ii)	• covers end (of ball) of humerus/L; • covers depression in ulna/M;		2
(c)	• movement not smooth/bones grind/rub together/less shock absorption/increased friction; • painful;		2
(d)	any four from • attached to end of muscle/attach muscle to bone; • inelastic/do not stretch; • muscle contracts; • pulls tendon; • causing M/ulna/bone to move;		4 Total 12

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 5 of 9 · Question 6 part d · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 6 of 9 · Question 3 part c · 5 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

number				
3	(a) (i)	A = synovial fluid; B = synovial membrane;		2
	(ii)	• lubricate joint; • for easier movement/to reduce friction;		2
	(b) (i)	• one section covering ball; • one section covering semi-circular part of bone; •		2
	(ii)	• movement not smooth/bones rub together/friction/less shock absorbing; • painful;		2
	(c)	• tendons attach to muscles; • attached to bone/lower arm; • muscle/biceps/triceps contracts; • generates pull on tendons/lower arm/bone; • tendons inelastic;	attaches muscles to bones = 2	5

Total : 13 marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 6 of 9 · Question 3 part c · 5 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 7 of 9 · Question 10 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number				
10	(a)	Any three from the following: • bones break more easily; • loss of height over time; • poor posture/difficulty walking/exercising; • (back) pain;		3
	(b)	Any three from the following: • muscles attached to bones; • (muscles) work antagonistically; • When one muscle contracts the other relaxes; • bone is pulled (to bring about movement);		3

Total marks 6

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 7 of 9 · Question 10 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Explain how biceps, triceps and tendons move the arm

Answer 8 of 9 · Question 1 part whole · 6 marks

2025 JUN · Paper 01 · find part whole in this mark scheme extract

1	• antagonistic; • hinge; • biceps; • triceps; • tendons; • cartilage;		6
			6 marks

Question number	Answer	Notes	Marks

Type MS · Explain how biceps, triceps and tendons move the arm

Answer 9 of 9 · Question 1 part b.i · 4 marks

2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

1 (a) (i) long (bone) — 1

(ii)

Part	Function: reduces friction	Function: produces red blood cells	Function: provides strength
compact bone			✓
spongy bone		✓	
cartilage	✓		

3

(iii)

- provides nutrients/named nutrient/oxygen;
- removes waste/named waste

2

(b) (i) In the following order:

- tendons;
- contracts;
- relaxes;
- pulls;

4

(ii) 0.5 x 125;
62.5 (J);

Accept 63 (J)

2

Total for Question 1 = 12 marks